Design, Modeling, and Simulation of a Quarter-Car Anti-Lock Braking System (ABS)

1. Executive Summary

This report presents the Model-Based Design (MBD), mathematical modeling, and simulation of a Quarter-Car Anti-Lock Braking System (ABS) implemented using MATLAB/Simulink.

The primary objective of the system is to maintain maximum tire traction during hard braking by regulating the tire slip ratio around its optimal reference value:

$$\lambda_{\text{ref}} = 0.20$$

By controlling wheel slip, the ABS prevent wheel lockup, reduces stopping distance, and helps preserve vehicle steerability and stability.

The simulation investigates the performance of the ABS under different braking conditions, including a comparison between ABS-enabled and conventional braking systems and a road-surface transition from dry asphalt to wet pavement.

2. System Architecture and Mathematical Modeling

2.1. Vehicle and Wheel Dynamics (Quarter-Car Model)

The quarter-car model represents the longitudinal motion of the vehicle and the rotational motion of a single wheel. These dynamics are described using Newton's laws of motion.

A. Vehicle Longitudinal Motion

The vehicle's longitudinal acceleration is calculated using:

$$\boxed{\dot{v} = -\frac{F_x}{m_q}}$$

Where:

- v: Vehicle longitudinal velocity (m/s).
- m_q: Quarter-car mass (350 kg).
- F_x: Longitudinal tire friction force (N).
- $\dot{v}$: Vehicle longitudinal acceleration (m/s²).

The negative sign indicates that the friction force acts against the vehicle's direction of motion during braking.

B. Wheel Rotational Motion

The rotational dynamics of the wheel are described by:

$$\boxed{\dot{\omega} = \frac{T_{\text{road}} - T_b}{J_w}}$$

Since the road torque is generated by the longitudinal tire force:

$$T_{\text{road}} = R_{\text{eff}} F_x$$

The rotational equation becomes:

$$\boxed{\dot{\omega} = \frac{R_{\text{eff}} F_x - T_b}{J_w}}$$

Where:

- ω: Wheel angular velocity (rad/s).
- J_w: Wheel rotational inertia (1.8 kg·m²).

- R_{eff}: Effective tire radius (0.30 m).
- T_b: Applied braking torque (N·m).
- T_{road}: Road torque acting on the wheel (N·m).

These equations describe how the braking torque and road friction influence the vehicle's deceleration and wheel rotation.

2.2. Tire Dynamics (Pacejka Magic Formula)

The longitudinal tire friction force is nonlinearly dependent on the wheel slip ratio. The Pacejka Magic Formula is used to describe the relationship between tire slip and road adhesion.

A. Slip Ratio Calculation

The braking slip ratio is defined as:

$$\boxed{\lambda = \frac{v - R_{\text{eff}}\omega}{\max(v, v_\epsilon)}}$$

Where:

$$v_\epsilon = 0.5\text{m/s}$$

The small velocity threshold v_ϵ prevents division by zero and numerical instability at very low vehicle speeds.

Under braking conditions:

- $\lambda = 0$: Pure rolling without longitudinal slip.
- $\lambda \approx 0.20$: Target operating slip for the ABS controller.
- $\lambda = 1$: Fully locked wheel.

B. Pacejka Magic Formula

The nonlinear tire-road friction coefficient is modeled using:

$$\boxed{\mu(\lambda) = D\sin[C\arctan(B\lambda - E(B\lambda - \arctan(B\lambda)))]}$$

The longitudinal tire force is then calculated as:

$$F_x = \mu(\lambda)F_z$$

Assuming a constant vertical load:

$$F_z = m_q g$$

Therefore:

$$\boxed{F_x = \mu(\lambda)m_q g}$$

Pacejka Model Parameters

Parameter	Symbol	Value
Stiffness factor	B	10
Shape factor	C	1.9
Peak friction factor (dry asphalt)	D	1.0
Curvature factor	E	0.97
Gravitational acceleration	g	9.81 m/s²

For a wet road surface, the peak friction factor is reduced to:

$$D_{\text{wet}} = 0.4$$

This represents a decrease in the available tire-road adhesion.

2.3. ABS Controller and Hydraulic Actuator

The ABS uses a closed-loop control system to regulate braking torque according to the difference between the actual and reference slip ratios.

A. Slip Ratio Error

The controller error is defined as:

$$\boxed{e(t) = \lambda(t) - \lambda_{\text{ref}}}$$

Where:

$$\lambda_{\text{ref}} = 0.20$$

The controller adjusts braking effort based on this error to maintain wheel slip close to the target.

B. Proportional-Integral (PI) Controller

The braking command is generated using a PI controller:

$$\boxed{u_b(t) = K_p e(t) + K_i \int_0^t e(\tau) d\tau}$$

The controller gains are:

$$K_p = 100$$
$$K_i = 10$$

The controller output is constrained using a saturation block:

$$\boxed{0 \le u_b(t) \le 1}$$

This ensures that the command remains within the allowable brake actuator range.

C. Hydraulic Brake Actuator Dynamics

The hydraulic brake actuator is modeled as a first-order system:

$$\boxed{G_a(s) = \frac{T_{\max}}{\tau_b s + 1}}$$

Where:

- $T_{\max} = 2000$N·m: Maximum braking torque.
- $\tau_b = 0.03$s: Hydraulic actuator time constant.
- s: Laplace-domain variable.

The actuator model represents the delay between the commanded brake input and the actual braking torque applied to the wheel.

---

3. Key Simulation Parameters

The main simulation parameters are summarized below.

Parameter	Symbol	Value	Unit
Quarter-car mass	m_q	350	kg
Wheel rotational inertia	J_w	1.8	kg·m²
Effective tire radius	R_{eff}	0.30	m
Initial vehicle velocity	v_0	27.78 (100 km/h)	m/s
Pacejka peak friction (dry)	D	1.0	—
Pacejka peak friction (wet)	D	0.4	—

Parameter	Symbol	Value	Unit
Pacejka stiffness factor	B	10	—
Pacejka shape factor	C	1.9	—
Pacejka curvature factor	E	0.97	—
Maximum braking torque	$T_{\max}$	2000	N·m
Actuator time constant	τ_b	0.03	s
Proportional gain	K_p	100	—
Integral gain	K_i	10	—
Reference slip ratio	λ_{ref}	0.20	—
Minimum velocity threshold	v_ϵ	0.5	m/s

4. Simulation Results and Analysis

4.1. Scenario 1: Performance Benchmark (ABS ON vs. ABS OFF)

The first simulation scenario evaluates braking performance with and without ABS control under dry-road conditions.

A. ABS OFF Mode

In conventional braking mode, the maximum braking torque is applied without slip regulation:

$$T_b = 2000\text{N·m}$$

According to the reported simulation results, the excessive braking torque causes the wheel to lock within approximately 0.2 seconds.

At full wheel lockup:

$$\omega = 0\text{rad/s}$$
$$\lambda = 1.0$$

The wheel stops rotating while the vehicle continues moving, resulting in sliding contact between the tire and the road.

The reported stopping distance is:

$$\boxed{d_{\text{ABS OFF}} = 43.5\text{m}}$$

B. ABS ON Mode

When ABS is enabled, the PI controller continuously adjusts the braking torque based on the measured slip ratio.

Instead of applying a constant maximum braking torque, the system modulates the torque around:

$$T_b \approx 1100\text{N·m}$$

The slip ratio remains approximately equal to its reference value:

$$\lambda \approx 0.20$$

This allows the wheel to continue rotating during braking while maintaining high tire-road friction.

The reported stopping distance is:

$$\boxed{d_{\text{ABS ON}} = 39.8\text{m}}$$

C. Braking Performance Improvement

The reduction in stopping distance is calculated as:

$$\Delta d = d_{\text{ABS OFF}} - d_{\text{ABS ON}}$$

Substituting the reported values:

$$\Delta d = 43.5 - 39.8$$

$$\boxed{\Delta d = 3.7\text{m}}$$

The percentage improvement is calculated as:

$$\text{Improvement} = \frac{\Delta d}{d_{\text{ABS OFF}}} \times 100$$

$$= \frac{3.7}{43.5} \times 100$$

$$\boxed{\text{Improvement} \approx 8.5\%}$$

Performance Comparison

Performance Metric	ABS OFF	ABS ON
Braking strategy	Constant maximum torque	Controlled braking torque
Brake torque	2000 N·m	Approximately 1100 N·m
Wheel behavior	Wheel lockup	Continuous wheel rotation
Slip ratio	1.0 after lockup	Approximately 0.20
Stopping distance	43.5 m	39.8 m
Braking control	Open-loop	Closed-loop

4.2. Scenario 2: Road Surface Transition (Dry-to-Wet Test)

The second simulation scenario evaluates the ability of the ABS controller to adapt to a sudden reduction in available road friction.

A. Test Conditions

Initially, the vehicle travels on dry asphalt with a peak friction parameter of:

$$D_{\text{dry}} = 1.0$$

At the specified transition time:

$$t = 1.0\text{s}$$

The road surface changes to wet pavement:

$$D_{\text{wet}} = 0.4$$

This transition represents a sudden decrease in tire-road adhesion.

B. Controller Response

Before the transition, the braking torque is approximately:

$$T_{b,\text{dry}} \approx 1100\text{N·m}$$

Following the reduction in friction, the ABS controller decreases the commanded braking torque toward approximately:

$$T_{b,\text{wet}} \approx 440\text{N·m}$$

The reduction in braking torque helps prevent excessive wheel slip caused by the lower available road friction.